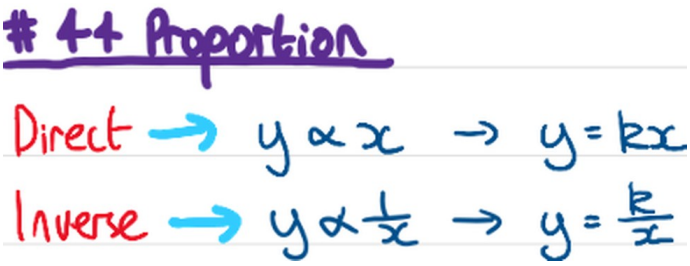
Worksheet 19 · Direct & Inverse Prop. IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Set up equations for direct proportion • Set up equations for inverse proportion • Use proportion formulae to find unknown values		
Grade: / 54	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes



Questions *Direct & Inverse Prop.*

Question 1 **(4 marks)**

T is directly proportional to the cube of r

$T = 21.76$ when $r = 4$

(a) Find a formula for T in terms of r

(3)

(b) Work out the value of T when $r = 6$

(1)

Question 2 **(4 marks)**

T is inversely proportional to m^2

$T = 30$ when $m = 0.5$

(a) Find a formula for T in terms of m.

(3)

(b) Work out the value of T when $m = 0.1$

(1)

Question 3 **(4 marks)**

M varies directly as the cube of h
$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

Question 4 (5 marks)

The following table gives values of x and y where y is inversely proportional to the square of x.

x	1.5	2	3	4
y	16	9	4	2.25

(a) Find a formula for y in terms of x.

.....

(3)

Given that $x > 0$

(b) find the value of x when $y = 144$

.....

(2)

Question 5 (5 marks)

y is directly proportional to the cube of x
$y = 20h$ when $x = h \quad (h \neq 0)$

(a) Find a formula for y in terms of x and h

$y =$ .....

(3)

(b) Find x in terms of h when $y = 67.5h$
Give your answer in its simplest form.

$x =$ .....

(2)

Question 6 (4 marks)

P is directly proportional to the cube of Q.
When $Q = 15$, $P = 1350$

(a) Find a formula for P in terms of Q.

$P =$ .....

(3)

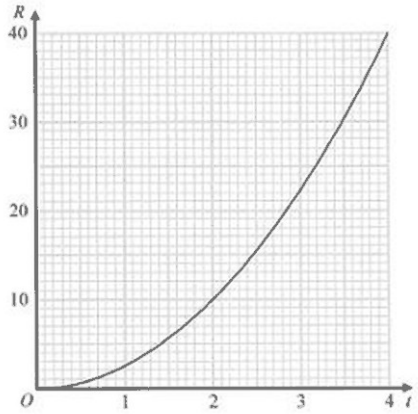
(b) Calculate the value of P when $Q = 20$

$P =$ .....

(1)

Question 7 (5 marks)

R is proportional to t^2
The graph shows the relationship between R and t for $0 \leqslant t \leqslant 4$



(a) Find a formula for R in terms of t.

iven also that $R = \frac{8}{5x}$

) show that t is inversely proportional to $\sqrt{x}$ for $t > 0$

Question 8 (5 marks)

P is inversely proportional to $\sqrt{q}$
$P = 10$ when $q = 0.0064$

(a) Find a formula for P in terms of q

.....

(3)

(b) Find the value of q when $P = 20$

.....

(2)

Question 9 (3 marks)

A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C =$ .....

Question 10 **(6 marks)**

A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

.....

(3)

(b) Find the value of A when $r = 7.5A$

$A =$ .....

(3)

Question 11 **(5 marks)**

M is inversely proportional to g^3
$M = 24$ when $g = 2.5$

(a) Find a formula for M in terms of g

.....

(3)

(b) Work out the value of g when $M = \frac{1}{9}$

$g =$ .....

(2)

Question 12 **(4 marks)**

y is inversely proportional to $\sqrt{x}$
x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T =$ .....